

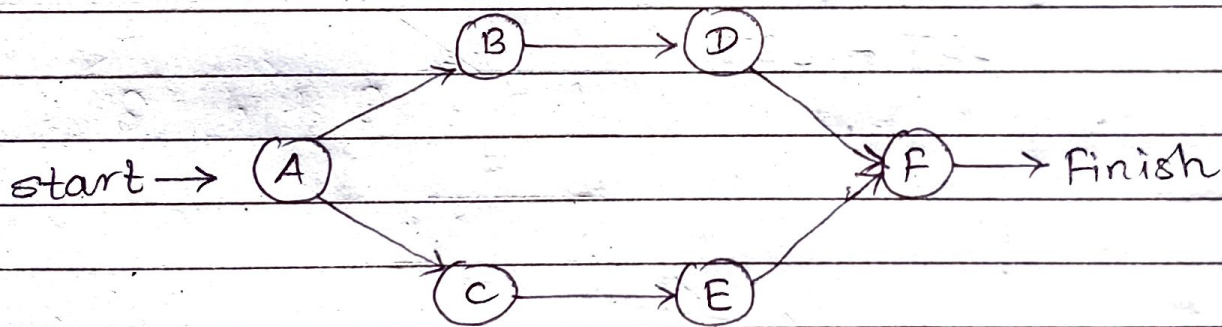
(1) Calculate the critical path and identify the project completion time of the following:

Activity	Predecessor	Duration
A	-	5
B	A	4
C	A	5
D	B	6
E	C	3
F	D, E	4

sol:

(i) Network diagram:

→ To draw network diagram, we use Activity on Node method as;



(ii) Calculate EST, LST, EFT, LFT and Float?

Let	EST	D	EFT
		Activity	
	LST	Float	LFT

where,

EST = Earliest start Time

LST = Latest start Time = ~~L~~EFT - Duration

EFT = Earliest finish Time = EST + Duration

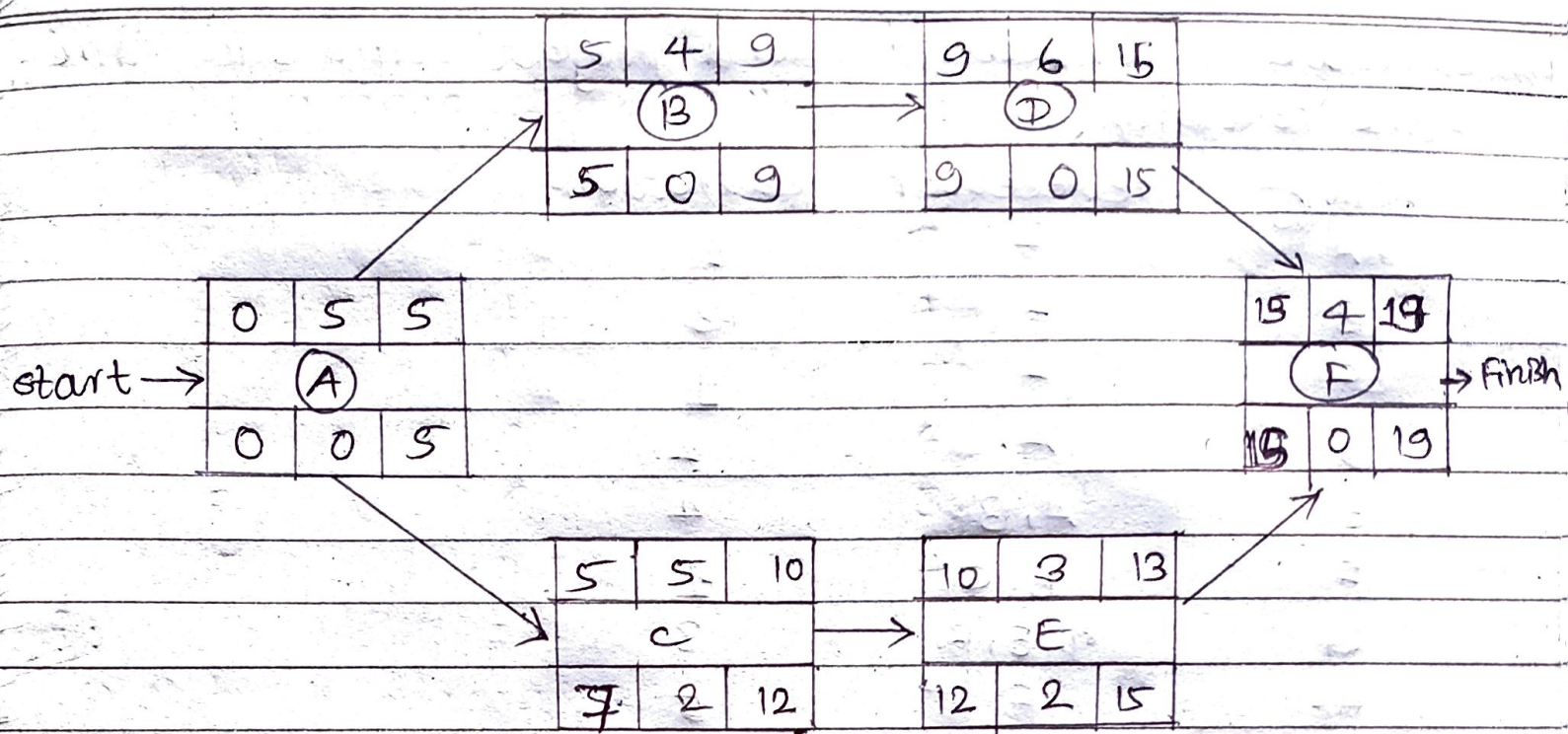
LFT = Latest finish Time

Activity = Given Activity

D = Duration

Float = ~~L~~LFT - EFT or

LST - EST



Hence, the required critical path is ~~AB~~ A → B → D → F.

and the project completion time is 19 days.

Ans.

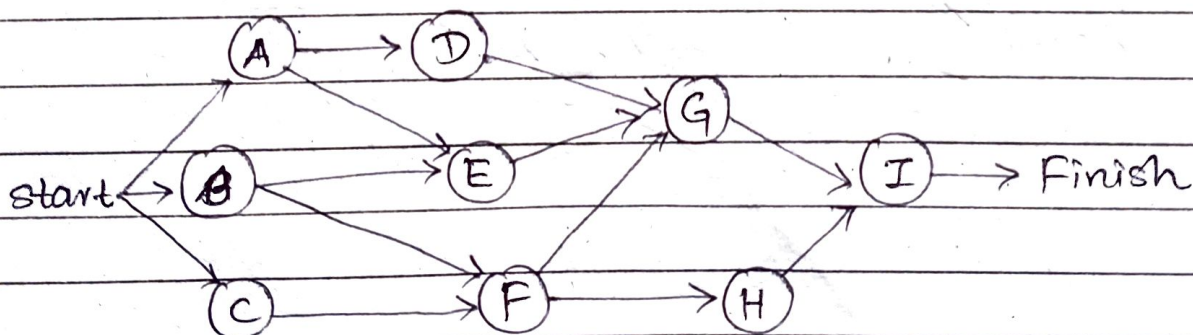
(2) Calculate the critical path and project completion time of the following:

Activity	Predecessor	Duration
A	-	2
B	-	3
C	-	1
D	A	2
E	A, B	4
F	B, C	1
G	D, E, F	3
H	F	5
I	G, H	3

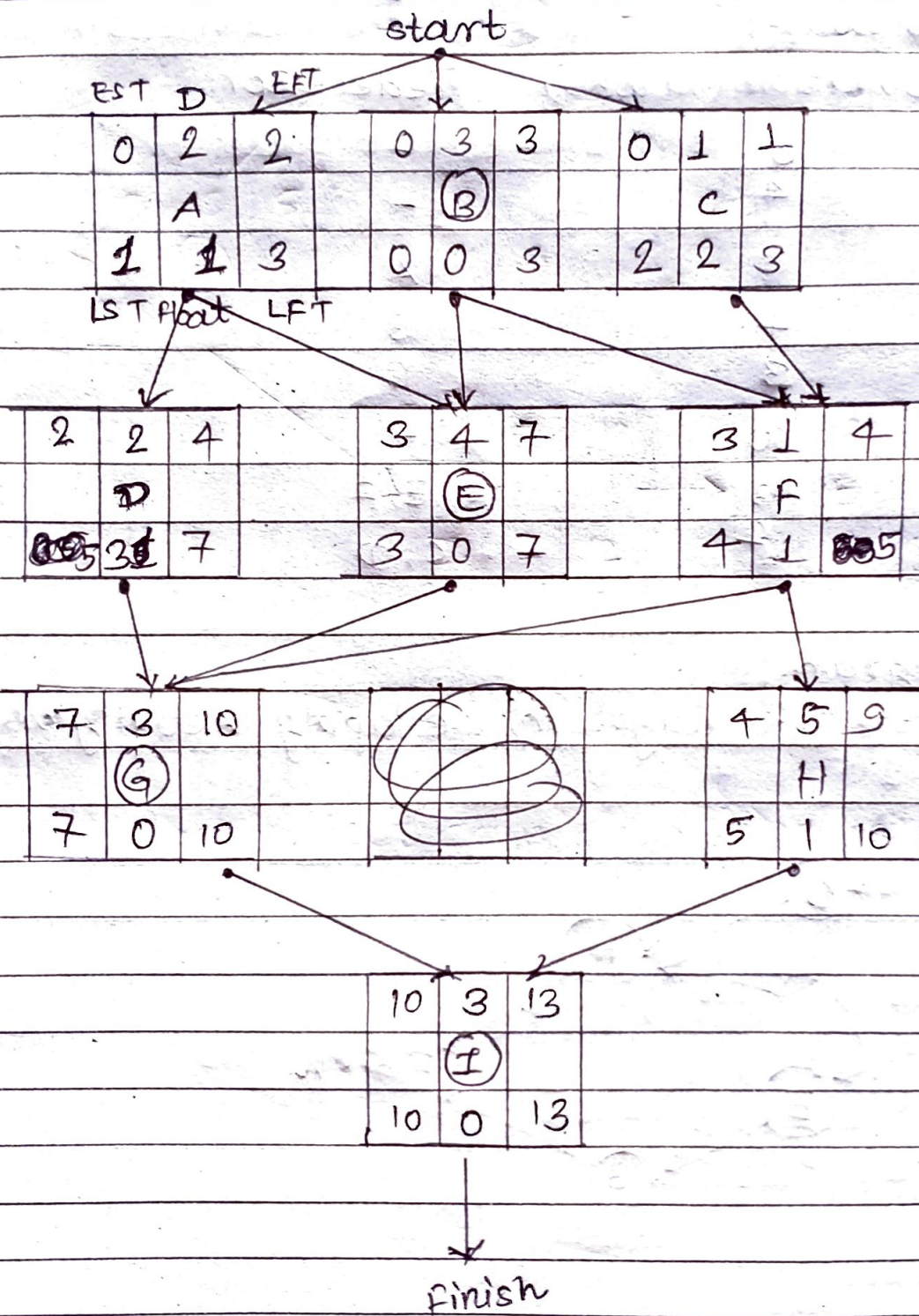
Sol:

(i) Draw Network diagram:

→ To draw network diagram, we apply Activity on Node method as shown below:



(ii) Calculate EST, LST, EFT and, LFT and Float:



Hence, the critical path is: $B \rightarrow E \rightarrow G \rightarrow I$.

and the project completion time is 13 days. Ans

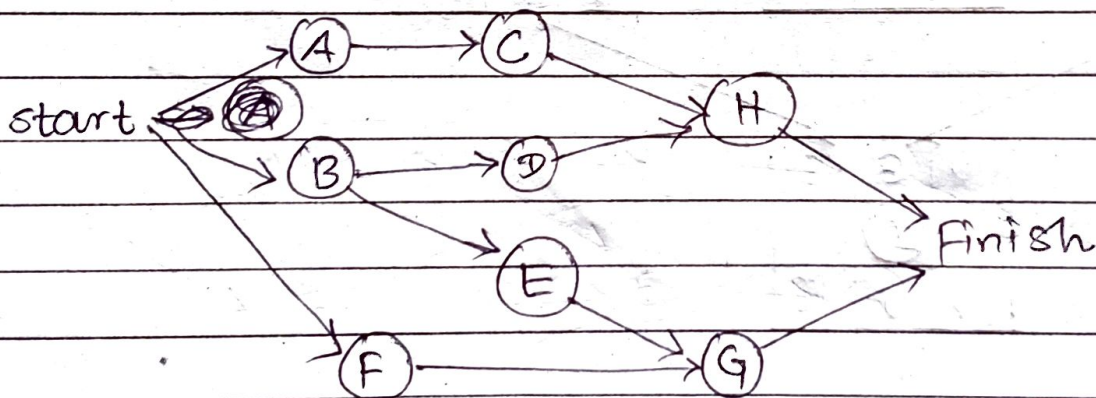
(3) Calculate the critical path and project completion time of the following:

Activity	Duration (in wks)	Predecessor
A	6	-
B	4	-
C	3	A
D	4	B
E	3	B
F	10	-
G	3	E, F
H	2	C, D

Sol:

(i) Network diagram:

→ To draw network diagram, we apply Activity on Node:



(ii) Calculate EST, LST, EFT, LFT and Float:

where,

EST	D	EFT
Activity		
LST	Float	LFT

EST = Earliest Start Time

D = Duration (Time)

EFT = Earliest finish Time

= EST + D

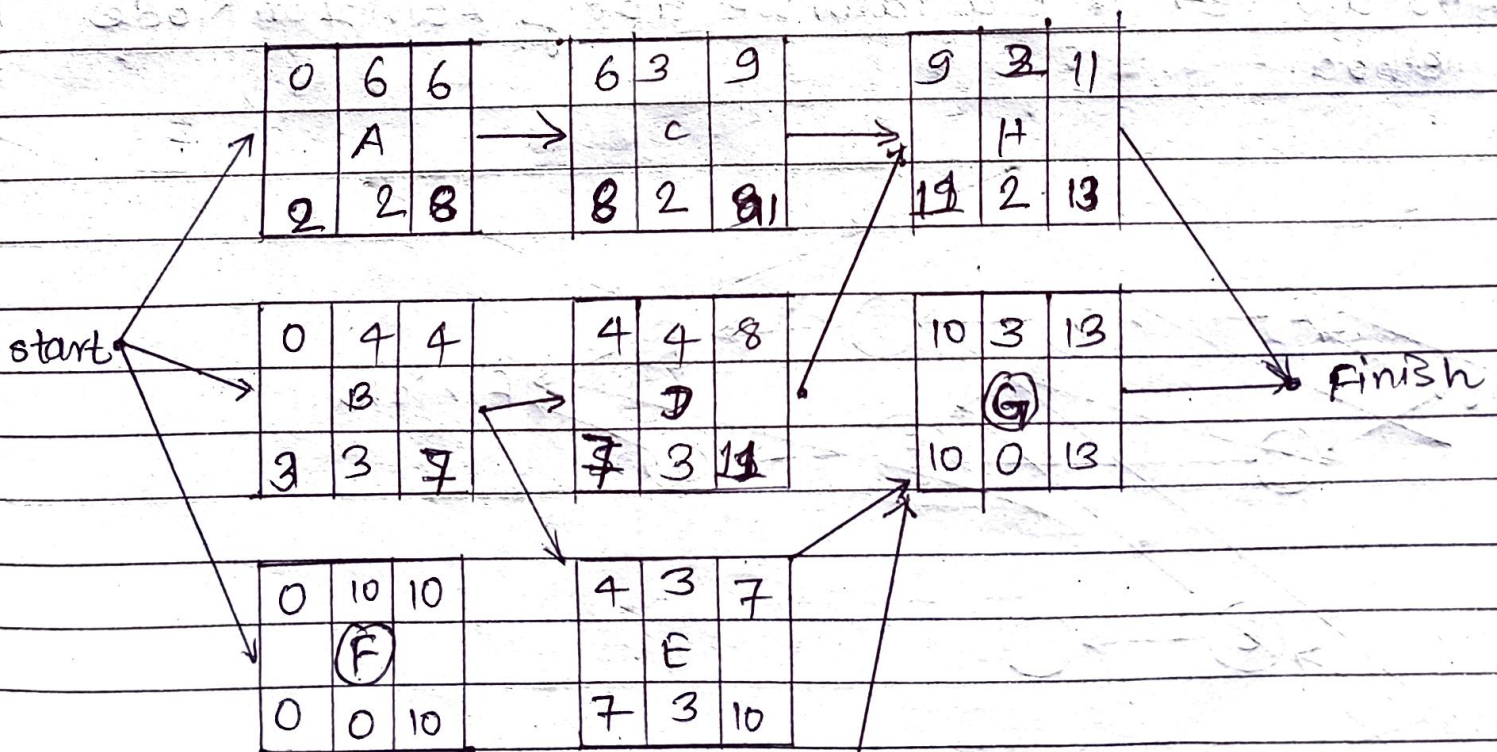
LFT = latest Finish Time

LST = latest start Time

= LFT - D

Float = EFT - EST or

LFT - LST



Hence, the required critical path is: F → G.

and the project completion time is 13 weeks. Ans

2022 - Fall

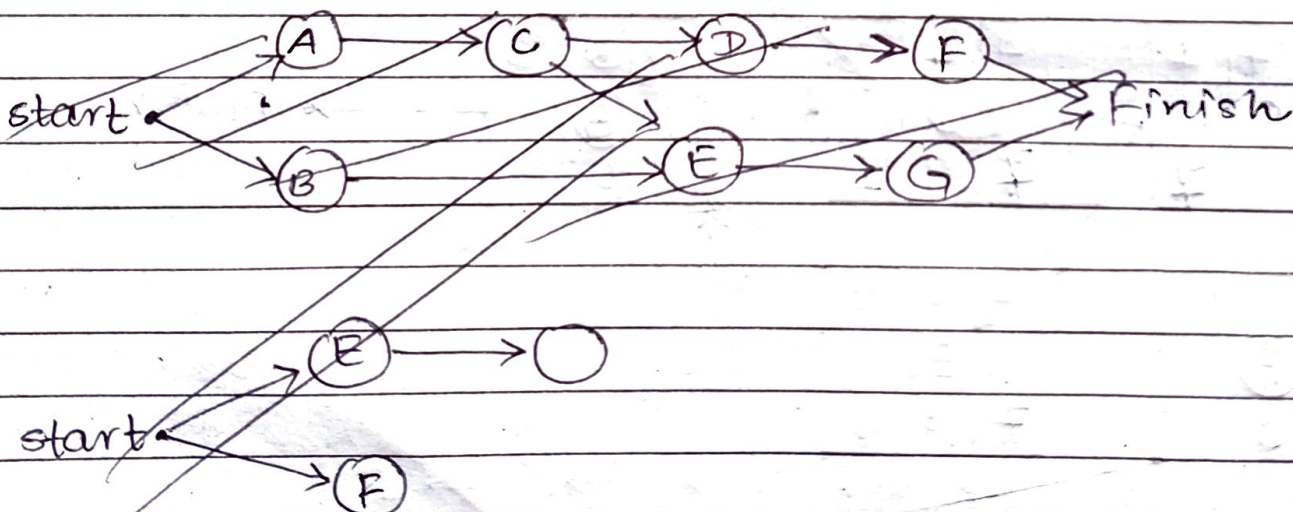
3.a) Draw Network diagram for the given set of activities and identify critical path and critical activities. Also calculate the project duration.

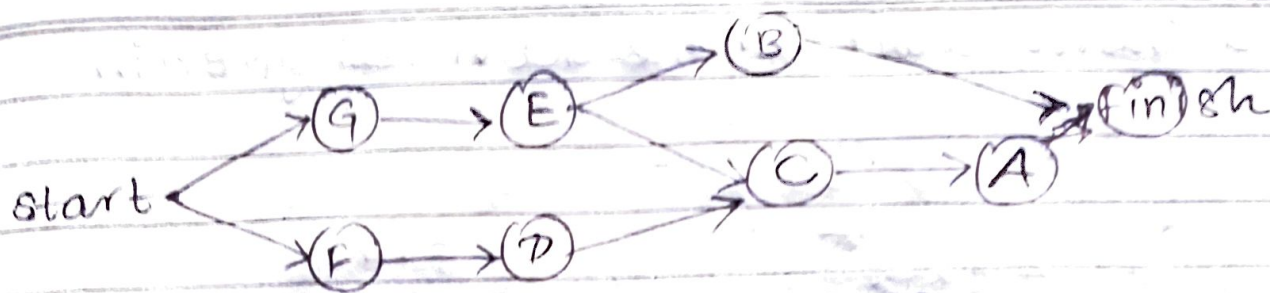
Activity	Duration	Immediate Successor
G	2	E
F	3	D
E	4	C, B
D	4	C
C	5	A
B	3	-
A	4	-

sol:-

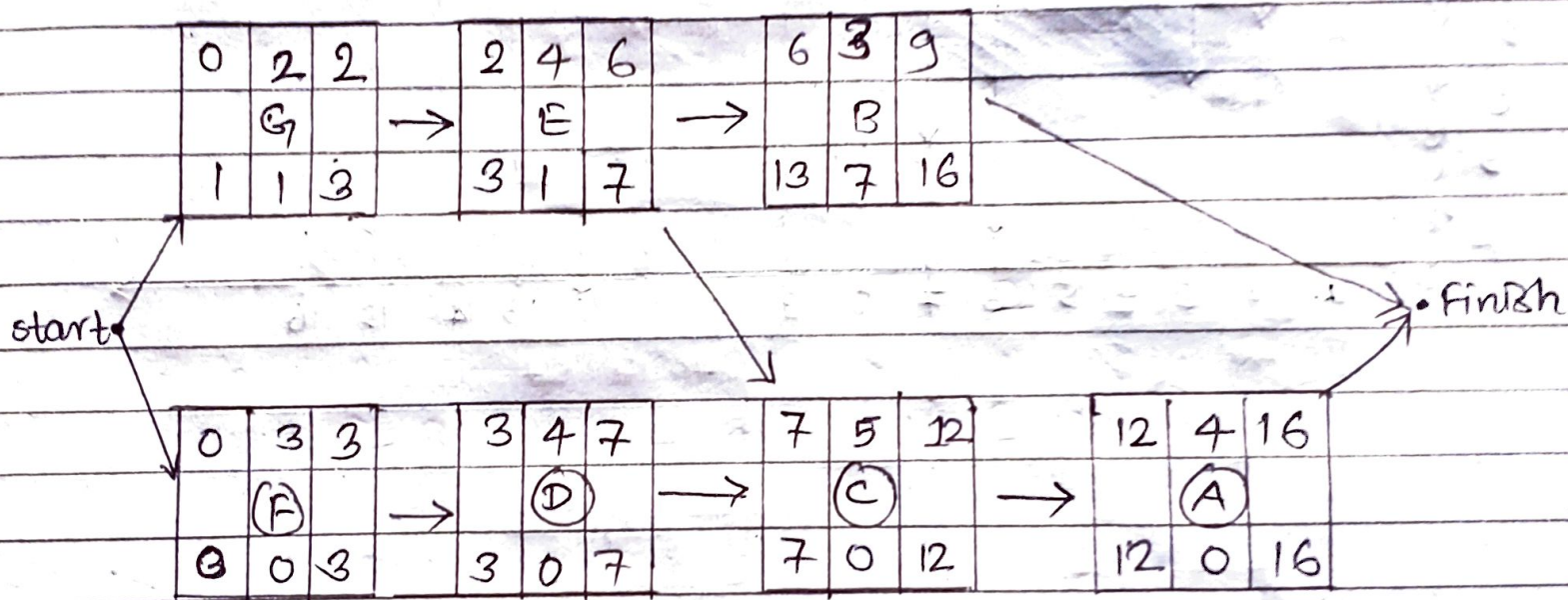
(i) Network diagram.

→ To draw network diagram, we apply Activity on Node method.





(ii) Calculate net EST, EFT, LST, LFT and Float:

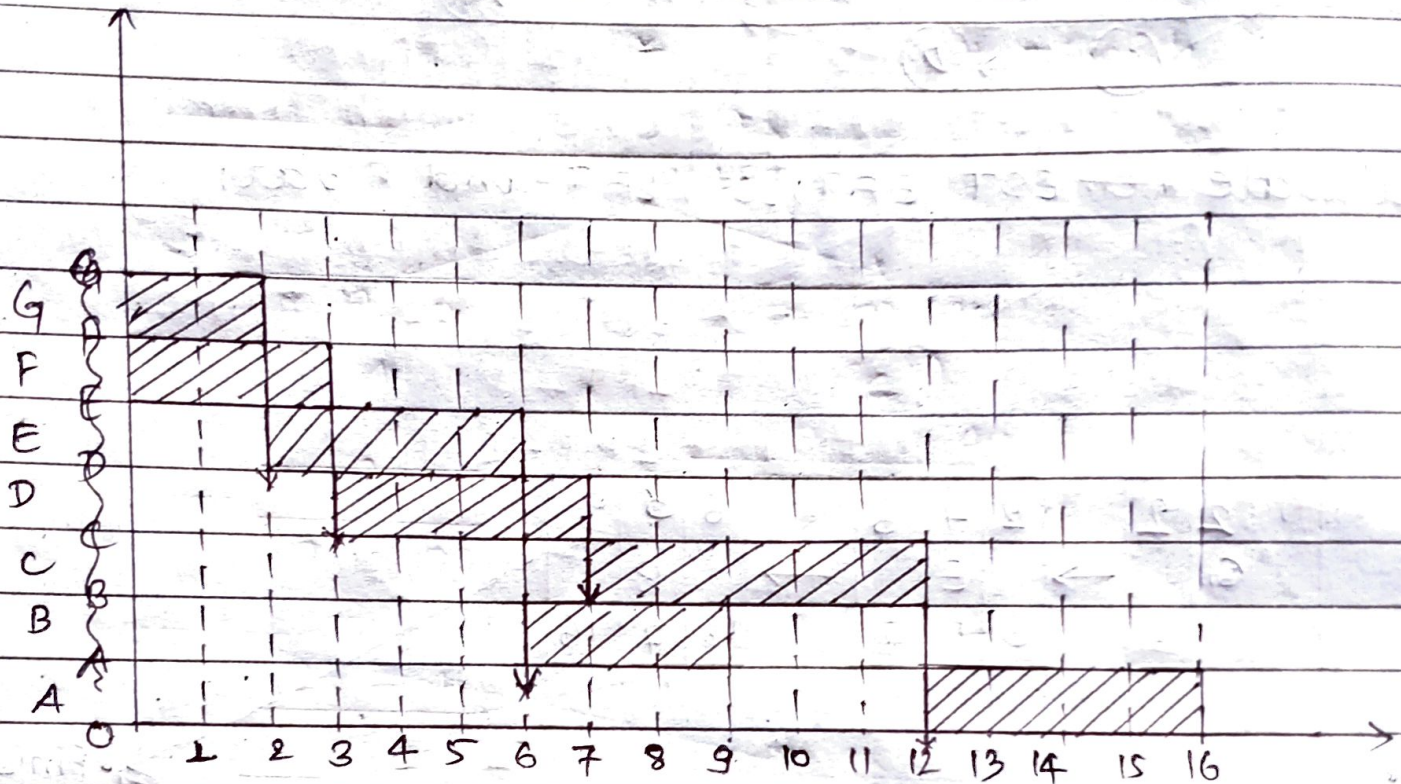


Hence, the project critical path is: $F \rightarrow D \rightarrow C \rightarrow A$.
and the critical activities are: F, D, C, A.

and the project duration is 16 days.

3.6) Draw a Gantt chart for the activities given in Q.3a).

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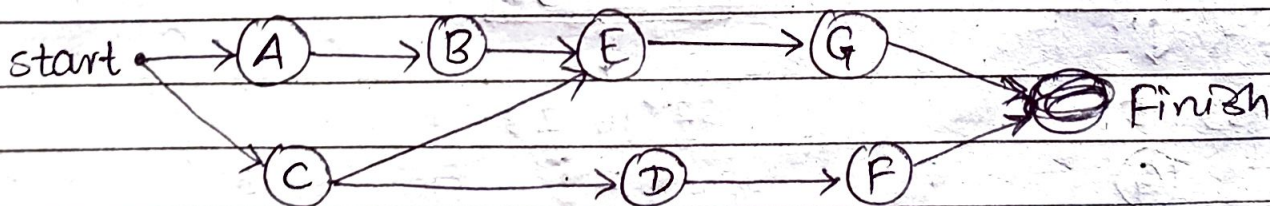


4a) Draw the network diagram for the given set of activities. Find out the estimated project duration and critical path. Also calculate total float and free float for each activity.

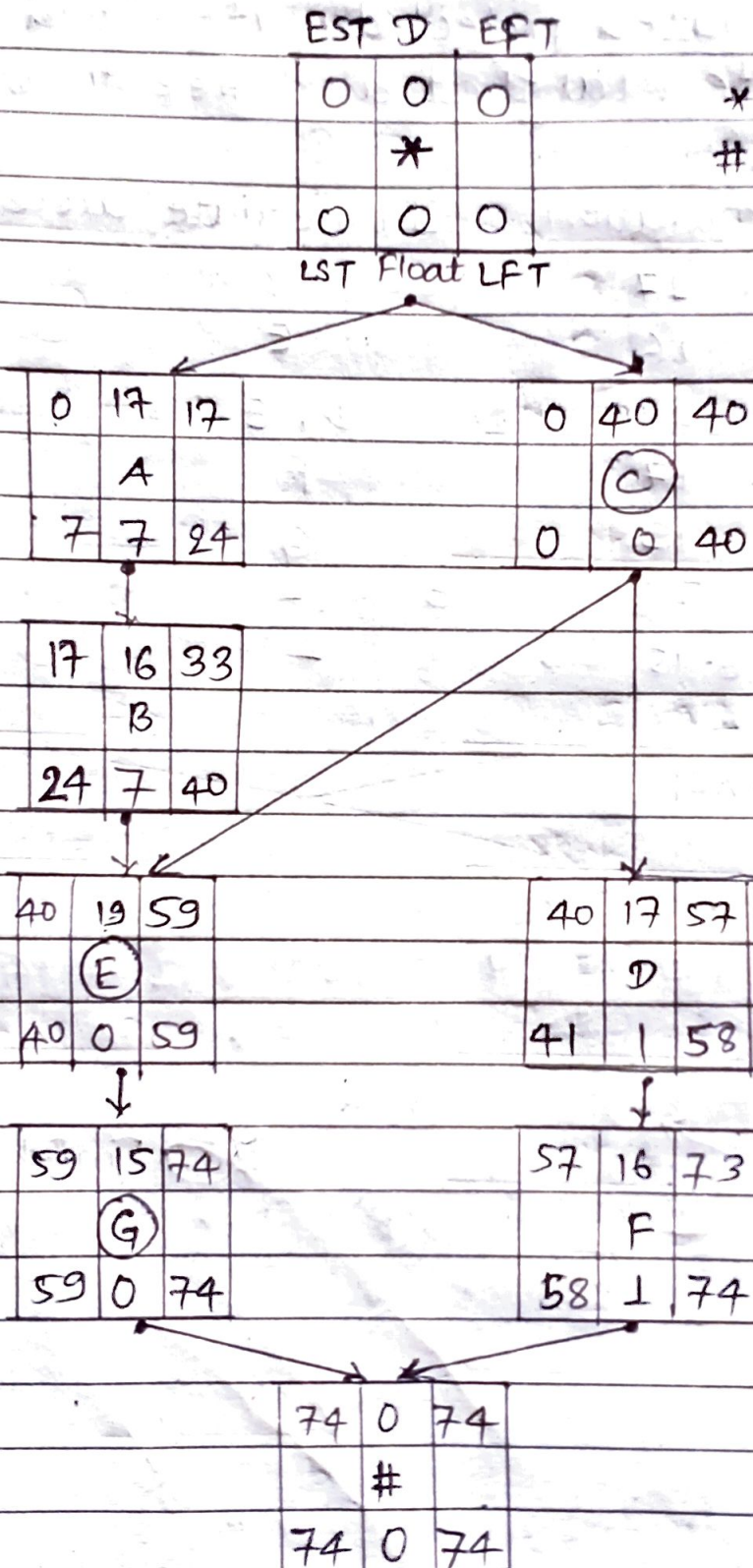
Activities	Duration	Immediate Successor
A	17	B
B	16	E
C	40	D, E
D	17	F
E	19	G
F	16	-
G	15	-

Q9

(i) Network diagram:



(ii) Calculate EST, LST, EFT, LFT and Float:



Hence, the critical path is $B \rightarrow E \rightarrow G$. AO
and project duration is 74 days AO

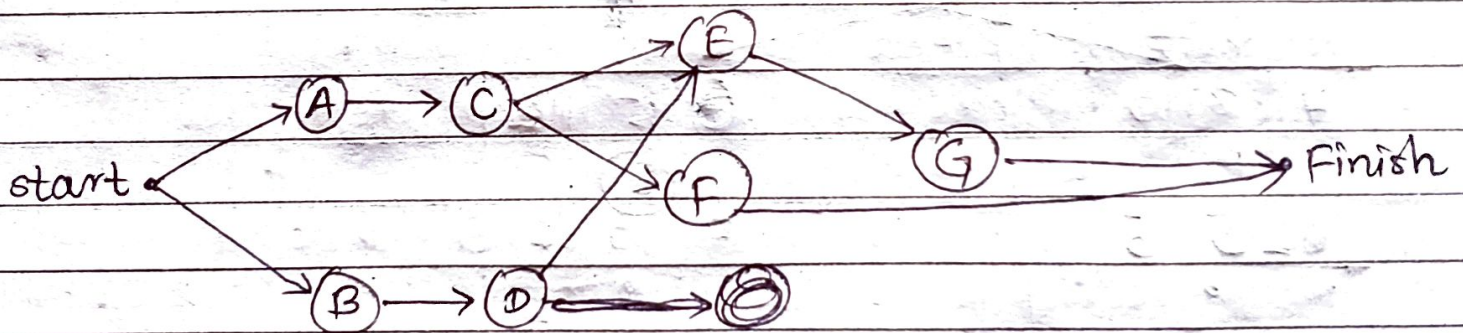
3.6) A project consists of the following activities.

Activity	Immediate Successor	Duration (Months)
A	C	2
B	D	3
C	E, F	4
D	E	4
E	G	5
F	G	3
G		4

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(i) Draw Network diagram:

→ To draw network diagram, we apply Activity on Node method.



(ii) Calculate EST, LST, EFT, LFT and Float:

EST	D	EFT
Activity		
LST	Float	LFT

EST = Earliest start Time

D = Duration

EFT = Earliest finish Time

~~EFT~~ = EST + D

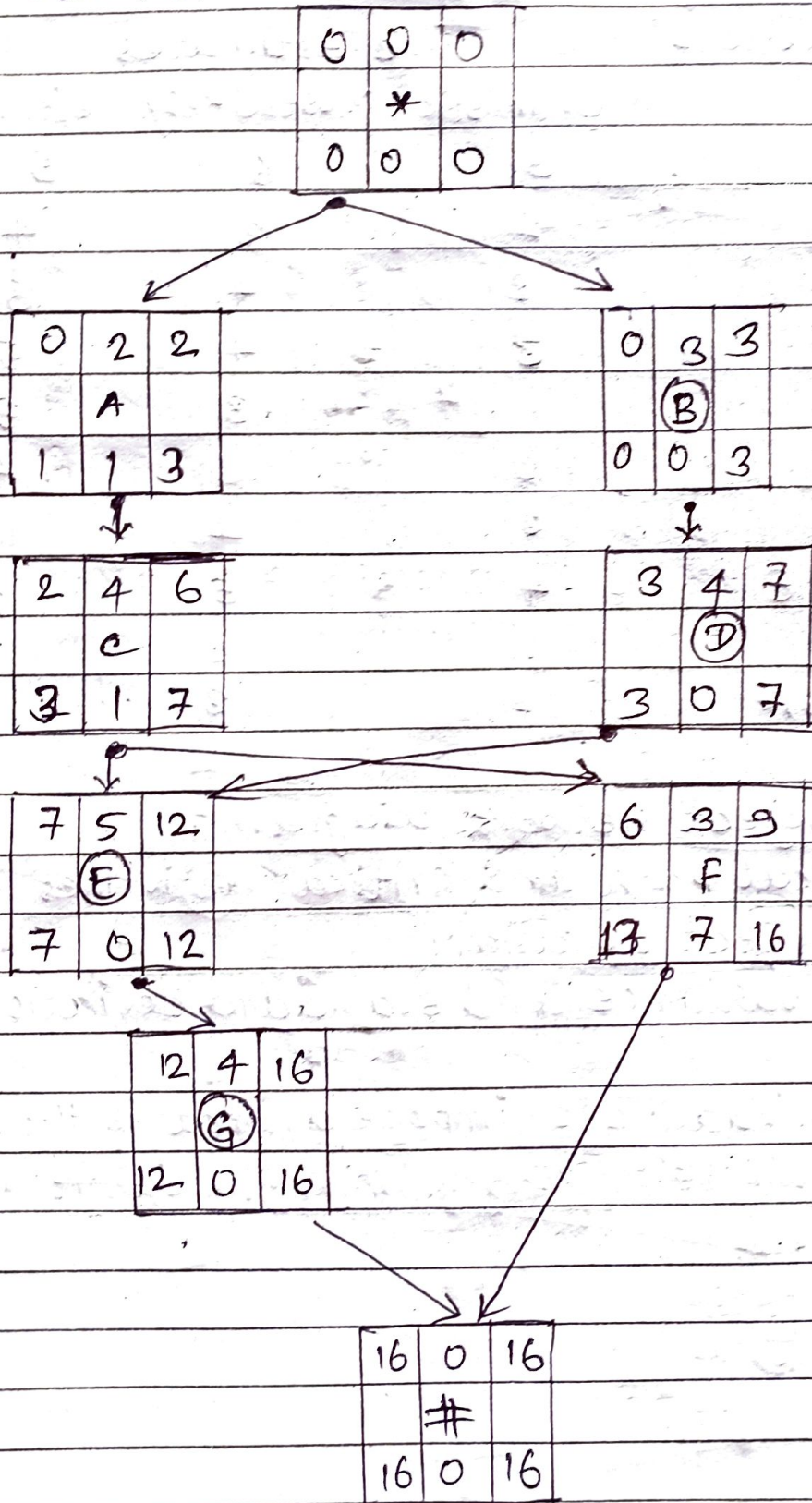
EFT = Earliest finish time

LST = Latest start Time

= LFT - D

Float = LFT - EFT or LST - EST

Let * = start & # = Finish;



Hence the critical path is: start → B → D → E → G → Finish
and the project duration is 16 days. Ans

4. b) A project consists of the following activities:

Activity	Predecessors	Time estimates		
		Optimistic	Most likely	Pessimistic
A	-	3	6	9
B	-	2	3	4
C	A	2	4	6
D	A	3	4	5
E	A	1	3	5
F	C	2	3	4
G	D	3	4	11
H	B, D, E	4	5	6
I	H	2	2	2
J	I, G, F	2	3	4

- i) ^{Draw} calculate a project network diagram.
- ii) Identify critical path and critical activities
- iii) calculate the project duration.
- iv) Calculate the variance and standard deviation of the project.
- v) If a 21 week deadline is imposed, what is the probability that the project will be finished within the limit time limit?

Sol

step 1: Calculate $t_e = \frac{t_o + 4t_m + t_p}{6}$ and Variance, σ^2

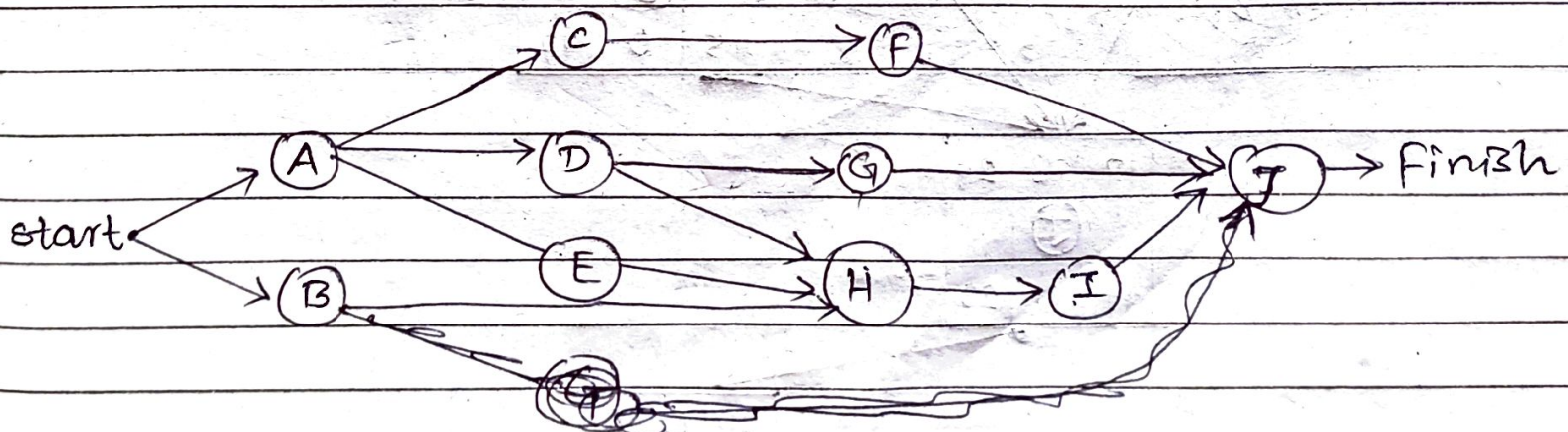
$$\sigma^2 = \left(\frac{t_p - t_o}{6} \right)^2$$

Now,

Activity	Predecessors	Time estimates			$t_e = \frac{t_o + 4t_m + t_p}{6}$	σ^2
		t_o	t_m	t_p		
✓ A	-	3	6	9	6	1
B	-	2	3	4	3	1/9
C	A	2	4	6	4	4/9
✓ D	A	3	4	5	4	1/9
E	A	1	3	5	3	4/9
F	C	2	3	4	3	1/9
G	D	3	4	11	5	16/9
✓ H	B, D, E	4	5	6	5	1/9
✓ I	H	2	2	2	2	0
✓ J	I, G, F	2	3	4	3	1/9

step 2: ~~To draw a~~ Draw network diagram:

To draw network diagram, we apply arrow activity on node method as;



Step 3: Calculate EST, EFT, LST, LFT and Float.

Let * = start

= finish

0	0	0
	*	
0	0	0

0	6	6
	A	
0	0	6

0	3	3
	B	
7		10

6	4	10
	C	
10	4	14

6	4	10
	D	
6	0	10

6	3	9
	E	
7	1	10

10	3	13
	F	
14	4	17

10	5	15
	G	
12	2	17

10	5	15
	H	
10	0	15

	I	

15	2	17
	J	
15	0	17

17	3	20
	K	
17	0	20

20	0	20
	#	
20	0	20

(ii) The critical path is:

Start $\rightarrow$ A $\rightarrow$ D $\rightarrow$ H $\rightarrow$ I $\rightarrow$ J $\rightarrow$ Finish

and the critical activities are: A, D, H, I, J.

(iii) The project duration is 20 ~~days~~ weeks. Ans

(iv)

$$\text{Variance, } \sigma^2 = 1 + \frac{1}{9} + \frac{1}{9} + 0 + \frac{1}{9} = \frac{4}{3}$$

$$\therefore \text{S.D., } \sigma = \sqrt{\sigma^2} = \sqrt{4/3} = 1.154$$

(v) Since 21 weeks deadline is imposed then the probability is

$$\left(\frac{21-20}{\text{S.D.}} \right) = \frac{-1}{1.154} = 0.866 \text{ i.e. } 86.6\% \quad \underline{\text{Ans}}$$

2017- Spring

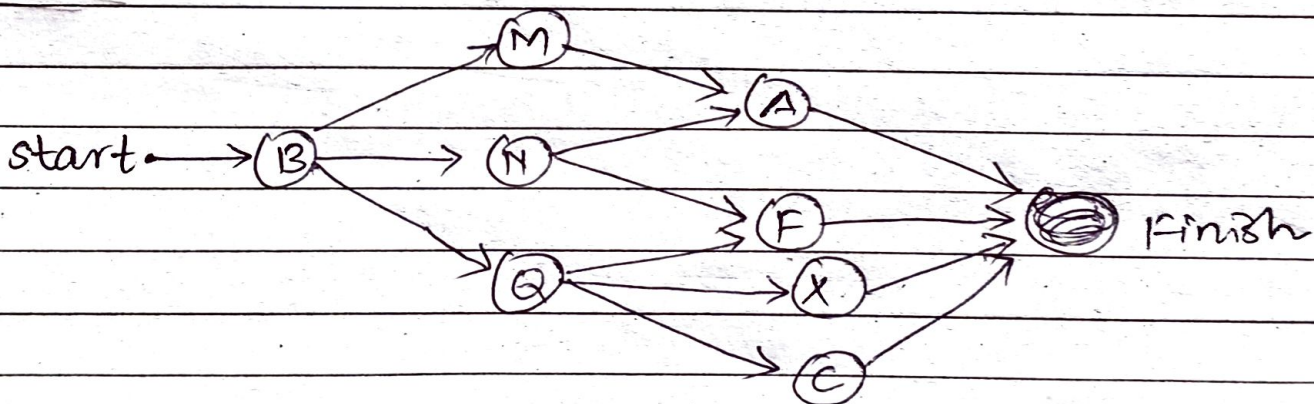
(b) Identify the critical path, Calculate EST, EFT, LST, LFT, and Float.

Activities	Duration	Predecessors
B	5	
M	4	B
N	9	B
Q	15	B
A	1	M, N
P	4	N, Q
X	9	Q
C	9	Q

809

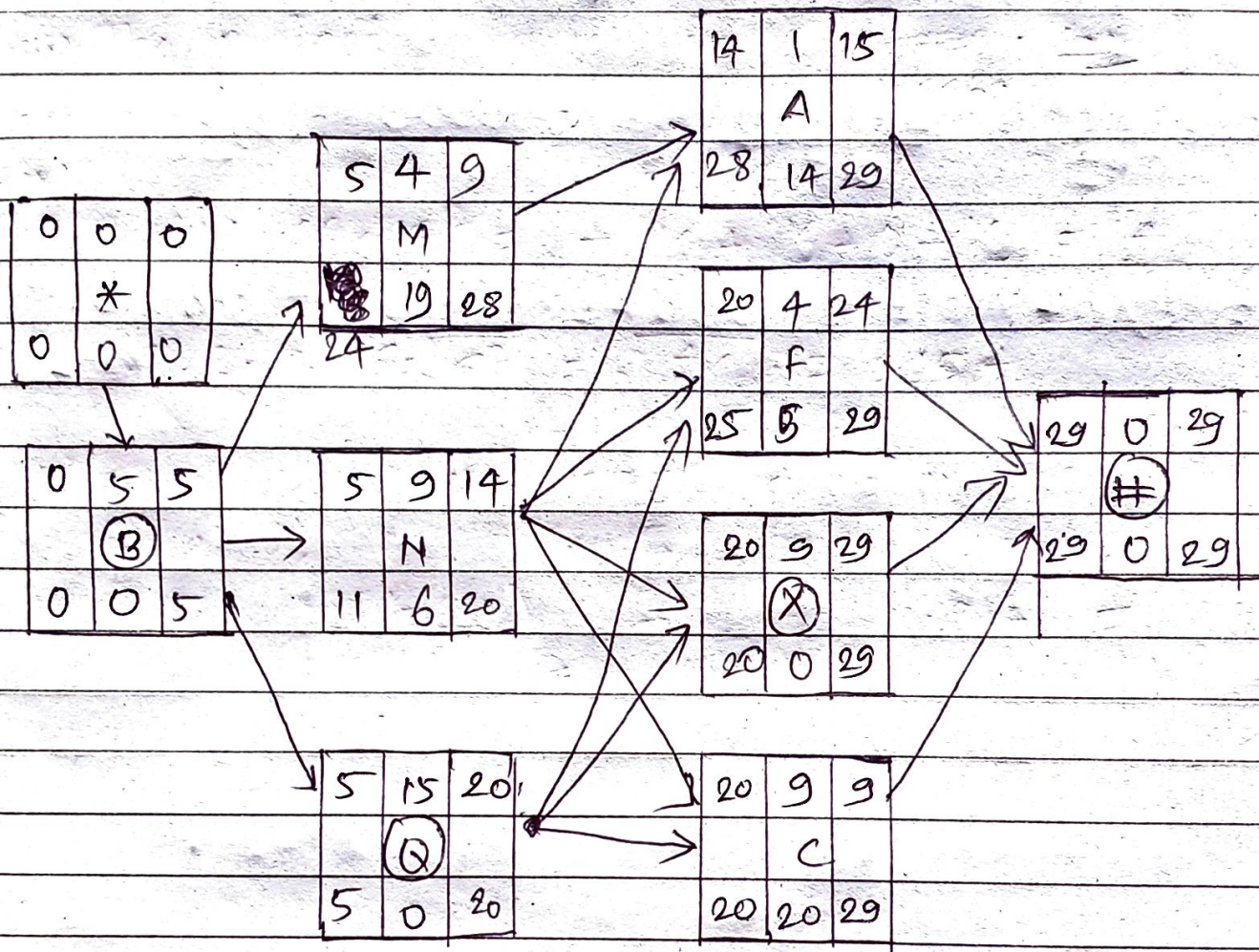
Step 1: Draw network diagram:

→ To draw network diagram, we apply activity on node method as:



step 2: Calculate EST, EFT, LST, LFT and Float.

Let * = start & # = finish.



Hence, the critical path is: ~~start~~

* → B → Q → X → #

and the critical activities are: B, Q, X.

Project Cost Estimation:

(1) Consider a project schedule to complete within 4 months with total estimated project cost of Rs. 4,00,000. At the end of 3 months Rs. 2,00,000 was spent while work done was of cost Rs. 1,00,000 with these dimension. Calculate CV, SV, CPI, SPI, EAC, ETC.

Given:

Budget At Completion, BAC = 4,00,000

Actual cost, AC = 2,00,000

Earned value, EV = 1,00,000

Planned Value, PV = $\frac{3 \times 4,00,000}{4} = 3,00,000$

$$(i) CV = EV - AC \text{ (cost Variance)} \\ = 1,00,000 - 2,00,000 = -100,000 < 0$$

i.e. the project is ~~under~~ budget
over the

$$(ii) SV = EV - PV \text{ (schedule Variance)} \\ = 1,00,000 - 3,00,000 \\ = -2,00,000 < 0 \text{ i.e. the project is behind the schedule.}$$

$$(iii) CPI = EV / AC \text{ (cost Performance Index)} \\ = 1,00,000 / 2,00,000 = 0.5 < 1$$

i.e. the project is over the budget.

(iv) $SPI = EV/PV$ (Schedule Performance Index)

$$= \frac{100000}{300000} = \frac{1}{3} < 1$$

i.e. the project is behind the schedule.

(v) $EAC = \frac{BAC}{CPI} = \frac{400000}{0.5} = 800000$ Ans

(vi) $ETC = EAC - AC = 800000 - 200000 = 600000$ Ans

(2) If EV is twice its AC of the project, calculate CPI & CV. Is the project over or under the budget?

sol Here,

$$\text{Let } AC = x$$

$$EV = 2x$$

$$(i) \text{ CPI} = \frac{EV}{AC} = \frac{2x}{x} = 2 > 1$$

$$(ii) \text{ CV} = EV - AC = 2x - x = x$$

$$\text{CV}(\%) = \left(\frac{EV - AC}{AC} \right) \times 100\% = \frac{x}{x} \times 100\% = 100\%$$

Since $CV > 0$ and $\text{CPI} > 1$, so the project is under project.

(3) If SPI is 0.75 with earned value being 60. Calculate planned value and also state whether the project is ahead or behind the schedule?

sol Given:-

$$SPI = 0.75$$

$$EV = 60$$

$$PV = ?$$

we know,

$$SPI = \frac{EV}{PV}$$

$$\Rightarrow PV = \frac{EV}{SPI}$$

$$\Rightarrow PV = \frac{60}{0.75}$$

$$\therefore PV = 80$$

And,

$$\text{Schedule Variance, } SV = EV - PV = 60 - 80 = -20.$$

Since $SV < 0$, so the projective is ~~over~~ the behind the schedule. Ans

python

(4) A project is scheduled for time of 12 months with estimated cost of 400000. After 3 months, evaluation is done and identified that 40% of work is accomplished with the cost of 200000. Calculate cost and ^{schedule} variance for the project.

sol Given;

$$\text{Estimated Cost} = 400000 = \text{BAC}$$

$$\text{Actual Cost, AC} = 200000$$

$$\text{Planned Value, PV} = \frac{3 \times 400000}{12} = 100000$$

$$\text{Earned Value, EV} = 40\% \text{ of } 400000$$

$$= 160000$$

$$(i) \text{ CV} = \text{EV} - \text{AC} = 1,60,000 - 200,000 = -40,000 < 0$$

i.e. the project is over budget. Ans

$$(ii) \text{ SV} = \text{EV} - \text{PV} = 1,60,000 - 1,00,000 = 60,000 > 0$$

i.e. the project is ahead ~~of~~ ^{the} schedule. Ans

2018 - Spring

5. b) Given the following information for one-year project, use earned value management (EVM) to calculate following for the project.

a. Cost Variance, CV

b. Schedule Variance, SV

c. Cost Performance Index, CPI

d. Schedule Performance Index, SPI

Given:-

Planned Value, PV = \$23,000

Earned Value, EV = \$2,000

Actual Cost, AC = \$25,000

Budget At Completion, BAC = \$1,20,000

$$\begin{aligned} \text{(a) Cost variance, CV} &= EV - AC = 2000 - 25000 \\ &= -23000 < 0 \end{aligned}$$

i.e. the project is ~~over~~^{over} budget.

$$\begin{aligned} \text{(b) Schedule Variance, SV} &= EV - PV = 2000 - 23000 \\ &= -21000 < 0 \end{aligned}$$

i.e. the project is behind the schedule.

c) Cost Performance Index, $CPI = \frac{EV}{AC}$

$$= \frac{2000}{25000} = 0.08 < 1$$

i.e. the project is over the budget.

d) Schedule Performance Index, $SPI = \frac{EV}{PV}$

$$= \frac{2000}{28000} = 0.07 < 1$$

i.e. the project is behind the schedule.

2017-Spring

8.6) Consider a project schedule to complete within 4 months with total estimated project cost of Rs. 8,00,000. At the end of 3 months, Rs. 4,00,000 was spent while work done was of cost Rs. 2,00,000. With these information, calculate CV, SV, CPI, SPI, EAC, ETC.

~~So~~ Given:-

Budget At Completion, BAC = 8,00,000

Earned Value, EV = 2,00,000

Actual Cost, AC = 4,00,000

Planned Value, PV = $3 \times 2,00,000 = 6,00,000$

$$(i) CV = EV - AC = 2,00,000 - 4,00,000 = -2,00,000 < 0$$

i.e. the project is over the budget.

$$(ii) SV = EV - PV = 2,00,000 - 6,00,000 = -4,00,000 < 0$$

i.e. the project is behind the schedule.

$$(iii) CPI = \frac{EV}{AC} = \frac{2,00,000}{4,00,000} = 0.5 < 1$$

i.e. the project is over the budget.

$$(iv) SPI = \frac{EV}{PV} = \frac{2,00,000}{6,00,000} = 0.33 < 1$$

i.e. the project is behind the schedule.

$$(v) EAC = \frac{BAC}{CPI} = \frac{8,00,000}{0.5} = 16,00,000$$

$$(vi) ETC = EAC - AC = 16,00,000 - 4,00,000$$

$$= 12,00,000$$

Ans

2015-Spring

3.a) Consider a project schedule to complete within 4 months with total estimated project cost of Rs 4,00,000. At the end of 3 months, 200,000 was spent while work done was of cost 1,00,000. With these information, Calculate CV, SV, CPI, SPI, EAC, ETC.

29 Given:-

Budget At Completion, BAC = 4,00,000

~~200,000~~ Earned Value, EV = 1,00,000

Actual Cost, AC = 2,00,000

Planned Value, PV = $\frac{3 \times 4,00,000}{4}$

= 3,00,000

$$(i) CV = EV - AC = 1,00,000 - 2,00,000 = -1,00,000 < 0$$

i.e. the project is over budget.

$$(ii) SV = EV - PV = 1,00,000 - 3,00,000 = -2,00,000 < 0$$

i.e.

i.e. the project is behind schedule.

$$(iii) CPI = \frac{EV}{AC} = \frac{1,00,000}{2,00,000} = 0.5 < 1$$

i.e. the project is over budget.

$$(iv) SPI = \frac{EV}{PV} = \frac{1,00,000}{3,00,000} = 0.33 < 1$$

i.e. the project is behind schedule.

$$(v) EAC = \frac{BAC}{CPI} = \frac{4,00,000}{0.5} = 8,00,000$$

$$(vi) ETC = EAC - AC = 8,00,000 - 2,00,000 = 6,00,000$$

NCIT

2024 ~~Spring~~ Fall

3.a)

(i) If SPI is 0.95 with earned value being 75. Calculate planned value and also state whether the project is ahead or behind schedule?

~~So~~ Given:-

$$SPI = 0.95$$

$$EV = 75$$

$$PV = ?$$

we know

$$SPI = \frac{EV}{PV}$$

$$\Rightarrow PV = \frac{EV}{SPI}$$

$$\Rightarrow PV = \frac{75}{0.95}$$

$$\therefore PV = 78.94 > 1$$

Since $PV > 1$, so the project is ~~at~~ ahead schedule.

(ii) A project is scheduled for time of 15 months with an estimated cost of 800000. After 3 months, the evaluation is done and identified that 48% of work is accomplished with cost of 500,000. Calculate cost and schedule variance for the project.

solⁿ Given:

Budget At Completion, BAC = 800,000

Actual Cost, AC = 500,000

Planned Value, PV = $3 \times \frac{800000}{15} = 1,60,000$

Earned value, EV = 48% of 800000
= 3,84,000

$$(i) CV = EV - AC = 3,84,000 - 500,000 = -1,16,000 < 0$$

i.e. the project is ~~under~~ over budget.

$$(ii) SV = EV - PV = 3,84,000 - 1,60,000 = 2,24,000 > 0$$

i.e. the project is ahead schedule.

United Technical College:

2024 - Fall

4. b) Given the following information for a one-year project, use the earned value management (EVM) method to calculate the following for the project.

- (i) Cost Variance, CV.
- (ii) Schedule Variance, SV.
- (iii) Cost Performance Index, CPI.
- (iv) Schedule Performance Index, SPI.
- (v) TCIP

809 ~~809~~ Given ;

Planned Value, PV = 23,000.00

Earned Value, EV = 2,000.00

Actual Value, AC = \$25,000.00

Budget At Completion, BAC = 1,20,000.00

$$(i) \text{ ~~CV~~ } CV = EV - AC = (2,000.00 - 25,000.00) \$ \\ = -23,000.00 \$ \quad \underline{\underline{Ans}}$$

$$(ii) SV = EV - PV = (2,000.00 - 23,000.00) \$ \\ = -21,000.00 \$ \quad \underline{\underline{Ans}}$$

$$(iii) CPI = \frac{EV}{AC} = \frac{2,000.00}{25,000.00} \\ = 0.08 \quad \underline{\underline{Ans}}$$

python

$$\begin{aligned} \text{(iv) SPI} &= \frac{EV}{PV} = \frac{\$2000.00}{\$23,000.00} \\ &= 0.0869 \quad \underline{\underline{Ans}} \end{aligned}$$

$$\begin{aligned} \text{(v) EAC} &= \frac{BAC}{CPI} = \frac{\cancel{0.08} \ 1,20,000.00}{0.08} \\ &= 15,00,000 \quad \underline{\underline{Ans}} \end{aligned}$$

$$\begin{aligned} \text{(vi) ETC} &= EAC - AC = 15,00,000 - 25,000 \\ &= 14,75,000 \quad \underline{\underline{Ans}} \end{aligned}$$

$$\begin{aligned} \text{(vii) TCPI} &= \frac{BAC - EV}{EAC - AC} \\ &= \frac{1,20,000 - 2000}{15,00,000 - 25,000} \\ &= 0.08 \quad \underline{\underline{Ans}} \end{aligned}$$

$$\begin{aligned} \text{(viii) VAC} &= BAC - EAC \\ &= 1,20,000 - 15,00,000 \\ &= -13,80,000 \quad \underline{\underline{Ans}} \end{aligned}$$

EST	D	EFT
Activity		
LST	Float	LFT

where,

EST = Earliest Start Time

D = Duration

EFT = Earliest Finish Time = EST + Duration

LST = Latest start Time

LFT = Latest Finish Time = LST + Duration

Float = LFT - LST or EFT - EST

Activity: Given Activity

$$t_e = \frac{t_o + 4t_m + t_p}{6}$$

$$\sigma^2 = \left(\frac{t_p - t_o}{6} \right)^2$$

$$\rightarrow CV = EV - AC$$

$$\rightarrow SV = EV - PV$$

$$\rightarrow CPI = \frac{EV}{AC}$$

$$\rightarrow SPI = \frac{EV}{PV}$$

$$\rightarrow EAC = \frac{BAC}{CPI}$$

$$\rightarrow ETC = \frac{EAC - AC}{AC}$$

$$\rightarrow TCPI = \frac{BAC - EV}{EAC - AC}$$

where,

CV = Cost Value Variance

SV = Schedule Value Variance

PV = Planned Value

EV = Earned Value

AC = Actual Cost

EPI = Cost Performance Index

SPI = Schedule Performance Index

BAC = Budget At Completion

EAC = Estimate At Completion

ETC = Estimate To Completion

$$\rightarrow BAC = L$$

$$\rightarrow VAC = BAC - EAC$$